Gauge spectral limits: finite lattices, coupling sets and the continuum target

The Hamiltonian route has distinct steps. T1 gives a unique physical vacuum and positive gap on each finite lattice. T2 supplies an existential volume-uniform strong-coupling gap for the Kogut–Susskind operator; explicit Wilson-transfer estimates concern a different operator. T2′ is the stronger all-coupling lower-bound property $\liminf_{N_s\to\infty}\delta(g;N_s)>0$, not an equivalence to absence of a bulk transition, and is not necessary at every bare coupling for a continuum limit. T3 requires controlled physical gap scaling along the weak-coupling trajectory, and T4 requires construction, nontriviality and the axioms. Spectral transfer between finite lattices and the specified limiting vacuum is an additional obligation. These latter steps remain open. All energies below are measured using $\hbar c/a$; volume and cutoff limits are separate.

1. The finite-lattice Hamiltonian

Let G be a compact connected simple Lie group with Lie algebra basis T^a normalized by $\operatorname{tr} T^aT^b=\frac12\delta^{ab}$ in a fixed faithful representation (for $SU(N)$ the fundamental). Let $\Lambda=(a\mathbb{Z}/N_sa\mathbb{Z})^3$ be the periodic spatial lattice with spacing a, side $L=N_sa$, link set $\mathcal{E}$ and plaquette set $\mathcal{P}$; $|\mathcal{E}|=3N_s^3$, $|\mathcal{P}|=3N_s^3$. The Hilbert space is $\mathcal{H}=L^2(G^{\mathcal{E}},\mathrm{Haar}^{\otimes\mathcal{E}})$. On the copy of G attached to a link ℓ, the electric field components E_ℓ^a are $-i$ times the left-invariant vector fields, and $\sum_a(E_\ell^a)^2=-\Delta_\ell$ is the Laplace–Beltrami operator of the bi-invariant metric, whose eigenvalues on the Peter–Weyl blocks are the quadratic Casimirs $C_2(R)$ with multiplicity $(\dim R)^2$. The Kogut–Susskind Hamiltonian (Phys. Rev. D 11 (1975) 395, metadata) in the convention used here is

$$H=\frac{\hbar c}{a}\Big[\frac{g^2}{2}\sum_{\ell\in\mathcal{E}}(-\Delta_\ell)+\frac{2}{g^2}\sum_{p\in\mathcal{P}}\big(N-\operatorname{Re}\operatorname{tr}U_p\big)\Big]=\frac{\hbar c}{a}\left[g^2K+g^{-2}V\right],$$

Here the plaquette trace constant N is the faithful representation dimension (the colour N for $SU(N)$), distinct from the number N_s of sites per side. The scaled operators are $K=\frac12\sum_\ell(-\Delta_\ell)$ and $V=2\sum_p(N-\operatorname{Re}\operatorname{tr}U_p)$. With g dimensionless, U_p the ordered product of the four link variables around p, and K, V the dimensionless electric and magnetic operators. Other conventions rescale g^2 by a fixed number. The dimensionless spectrum of $aH/(\hbar c)$ depends on g, N_s and G only:

$$\Delta_{a,L}=\frac{\hbar c}{a}\,\delta(g;N_s,G).$$

Gauge transformations $\gamma:\Lambda\to G$ act by $(\mathcal{V}_\gamma\psi)(U)=\psi(\gamma_xU_\ell\gamma_y^{-1})$ for $\ell=(x,y)$; they are unitary, commute with H (the Laplacian is bi-invariant and $\operatorname{tr}U_p$ is conjugation invariant) and preserve positivity of functions. The physical space $\mathcal{H}_{\mathrm{phys}}$ is the joint fixed space (Gauss's law).

2. T1: every finite lattice has a gap above a unique physical ground state

Theorem 1. For every $a>0$, $N_s\ge2$, $g>0$ and G as above:

1. H is self-adjoint, bounded below, and has compact resolvent; its spectrum is a sequence $E_0<E_1<\cdots\to\infty$ of eigenvalues of finite multiplicity.
2. E_0 is simple with a strictly positive eigenfunction ψ_0, so $\Delta_{a,L}=E_1-E_0>0$.
3. $\psi_0\in\mathcal{H}_{\mathrm{phys}}$, $\mathcal{H}_{\mathrm{phys}}$ reduces H, and the physical gap satisfies $\Delta_{a,L}^{\mathrm{phys}}\ge\Delta_{a,L}>0$.

Proof. (1) $-\Delta_\ell$ is essentially self-adjoint on $C^\infty(G)$, nonnegative, with compact resolvent, since G is a compact manifold and Peter–Weyl gives the eigenvalues $C_2(R)$ with finite multiplicities and $C_2(R)\to\infty$ along any enumeration of irreducible representations. $K=\frac12\sum_\ell(-\Delta_\ell)$ acts on the finite tensor product with eigenvalues $\frac12\sum_\ell C_2(R_\ell)$; for each bound E only finitely

many assignments $\ell \mapsto R_\ell$ have $\sum_\ell C_2(R_\ell) \le E$, so K has compact resolvent. V is multiplication by a continuous function with $0 \le V \le 4N|\mathcal{P}|$, hence bounded, and a bounded perturbation of an operator with compact resolvent has compact resolvent by the resolvent identity $(H-z)^{-1} = (H_0-z)^{-1}[1+(g^{-2}V)(H_0-z)^{-1}]^{-1}$ for z far in the negative axis. Discreteness with finite multiplicities and $E_n \to \infty$ follow, and E_0 is an eigenvalue.

(2) The heat kernel of $-\Delta$ on a compact connected Riemannian manifold is strictly positive for $t > 0$ (standard, from the strong maximum principle or the Peter–Weyl expansion; metadata level), so the kernel of e^{-tK} on $G^{\mathcal{E}}$, a product of strictly positive kernels, is strictly positive. With Brownian motion X_t on $G^{\mathcal{E}}$ generated by $-K$ (time rescaled by $g^2\hbar c/a$), the Feynman–Kac formula gives

$$(e^{-tH}\psi)(U) = \mathbb{E}^U\Big[\exp\Big(-\frac{\hbar c}{ag^2}\int_0^t V(X_s)\,ds\Big)\psi(X_t)\Big],$$

valid because V is bounded and continuous. For $\psi \ge 0$, $\psi \ne 0$, the weight is at least $e^{-t\hbar c\cdot 4N|\mathcal{P}|/(ag^2)} > 0$ and the law of X_t has a strictly positive density, so $(e^{-tH}\psi)(U) > 0$ for every U: e^{-tH} is positivity improving. Reed–Simon IV, Theorem XIII.44 (metadata; number corroborated as in B78) then gives that E_0 is simple with $\psi_0 > 0$ almost everywhere, and $E_1 > E_0$ because E_1 is the next eigenvalue of a discrete spectrum.

(3) $\mathcal{V}_\gamma\psi_0$ is a normalized eigenfunction for E_0, hence $\mathcal{V}_\gamma\psi_0 = c(\gamma)\psi_0$ with $|c| = 1$; both functions are positive, so $c(\gamma) = 1$ and $\psi_0 \in \mathcal{H}_{\text{phys}}$. The projection $P = \int \prod_x d\gamma_x\, \mathcal{V}_\gamma$ onto $\mathcal{H}_{\text{phys}}$ commutes with $(H-z)^{-1}$, so $\mathcal{H}_{\text{phys}}$ reduces H and the restricted operator has compact resolvent, ground state ψ_0, and next eigenvalue at least E_1. □

Theorem 1 defines the object whose limits the conjecture is about. It proves nothing about those limits: $\Delta_{a,L}$ is $\hbar c/a$ times a number that may tend to zero as $N_s \to \infty$ or as $g \to 0$.

3. T2: strong coupling, and why volume uniformity is the whole difficulty

At $g \to \infty$ the operator g^2K dominates. Its ground state is the constant function, with eigenvalue 0, and its excitations are link representations. Gauge invariance excludes a single excited link: the lowest physical excitation of K for a periodic $SU(N)$ lattice with side at least four is a four-link plaquette loop in the fundamental representation, with electric energy $4\cdot\frac{1}{2}C_2(R_{\min})\,g^2\hbar c/a = 2C_2(R_{\min})g^2\hbar c/a$; for $SU(N)$, $C_2(\text{fund}) = (N^2-1)/(2N)$. So the unperturbed physical gap is

$$\gamma_{\text{phys}}(g) = 2C_2(R_{\min})\,g^2\,\frac{\hbar c}{a},$$

uniform in the side length under this restriction. At shorter periodic sides winding cycles can be shorter; nonabelian networks can also branch.

The naive bound fails. The magnetic term satisfies $0 \le g^{-2}V \le 4N|\mathcal{P}|g^{-2}$ in operator norm, so positivity and min–max give $\Delta^{\text{phys}} \ge \gamma_{\text{phys}} - \|g^{-2}V\|\hbar c/a = (2C_2g^2 - 4N|\mathcal{P}|g^{-2})\hbar c/a$, positive only when $g^4 > 2N|\mathcal{P}|/C_2$. The bound is extensive in the volume and useless as $N_s \to \infty$. Volume uniformity therefore requires using that V is a sum of $|\mathcal{P}|$ terms each of norm at most $4Ng^{-2}$ and each acting on four links: locality, not smallness of the norm.

What is established. Osterwalder and Seiler (Ann. Phys. 110 (1978) 440; abstract, B78) prove for the Euclidean lattice theory with any compact gauge group, at sufficiently strong coupling, reflection positivity, the existence of the infinite-volume limit and Wilson's confinement bound by a cluster expansion; exponential clustering of gauge-invariant correlations uniform in the volume, that is a mass gap of the transfer matrix in units of $1/a$, is the standard consequence of the same expansion (not read at section level). Hamiltonian versions rest on gap-stability theorems for local perturbations of commuting, frustration-free Hamiltonians (Yarotsky, Commun. Math. Phys. 261 (2005) 799; Bravyi–Hastings–Michalakis, J. Math. Phys. 51 (2010)

093512; both metadata): g^2K is a sum of commuting link terms with a product ground state and a uniform local gap, and $g^{-2}V$ is a sum of bounded local terms. Yarotsky permits the infinite-dimensional local spaces and the relevant bounded perturbation. The full hypothesis check, with existential constants, is in the strong-coupling note. The finite-volume physical loop estimate above does not itself give a volume-uniform threshold.

4. Coupling sets, finite volume and spectral limits

Define, for the Kogut–Susskind Hamiltonian of this note, with gauge group G, lattice spacing a and N_s sites per side,

$$c(g) = \liminf_{N_s\to\infty} \delta(g; N_s, G), \qquad \mathcal{G} = \{g > 0 :\ c(g) > 0\}, \qquad \Delta_{a,L} = \frac{\hbar c}{a}\delta(g; N_s, G), \quad L = aN_s.$$

Here $c(g)$ is a dimensionless lower-gap function; the constant c in $\hbar c/a$ is the speed of light. The volume limit is taken at fixed a and g. To retain the obligations map's convention, δ is the gap on the **full link Hilbert space**. Its restriction to the Gauss-law space has a gap $\delta^{\text{phys}} \geq \delta$; closing of δ need not even close δ^{phys}. An analogous gapped set can be defined using δ^{phys}, but the two sets have not been identified here.

T1 gives positivity at every finite volume; T2 gives $[g_0(G), \infty) \subset \mathcal{G}$; and T2′, as an all-coupling finite-volume lower-bound statement, is exactly $\mathcal{G} = (0, \infty)$. Proposition 1 proves finite-volume continuity. Proposition 2 identifies the slowest temporal correlation decay with the spectral threshold **assuming a specified limiting vacuum theory**; it does not identify that threshold with $c(g)$. Global openness and closedness of $\mathcal{G}$ remain open. If openness is established, closedness and the known nonemptiness would imply T2′ by connectedness. This is a sufficient route to a lattice statement, not an equivalence with the continuum mass-gap conjecture or with phase-transition absence.

4a. Finite-volume continuity

Proposition 1. For fixed a, $N_s \geq 2$ and a nontrivial compact connected Lie group G, the map $g \mapsto \delta(g; N_s, G)$ is continuous and strictly positive on $(0, \infty)$, and locally piecewise real-analytic. The same holds for the physical gap.

Proof. Write

$$h(g) := \frac{aH(g)}{\hbar c} = \frac{g^2}{2}K_0 + \frac{2}{g^2}V_0,$$

with $K_0 = \sum_\ell(-\Delta_\ell) \geq 0$ having compact resolvent and $V_0 = \sum_p(N - \operatorname{Re}\operatorname{tr} U_p)$ bounded, with $0 \leq V_0 \leq 2N|\mathcal{P}|$ for $SU(N)$ in the fundamental representation. For a general compact connected group, use the fixed faithful representation and replace N by its dimension. In a complex neighbourhood of any $g > 0$, this is a holomorphic family of type (A) on the common domain $D(K_0)$. For real g it is self-adjoint with compact resolvent. The local analytic eigenvalue-branch theorem for a self-adjoint analytic family (Kato/Rellich analytic perturbation theory, standard theorem used here) supplies finitely many real-analytic branches in an isolated spectral cluster. The ground branch is simple by T1. In a neighbourhood of each g, the first excited eigenvalue is the minimum of the finitely many branches in its cluster, hence continuous. Distinct analytic branches have isolated crossings locally; identically coincident branches do not affect this conclusion. Thus the gap is locally piecewise analytic and continuous. T1 supplies its strict positivity. The gauge projection reduces both K_0 and V_0, so the same argument applies on the fixed physical subspace. □

Correction (2026-10-02). The original min–max explanation did not prove continuity: an infimum over subspaces of continuous functions need not be continuous. The analytic-branch argument above replaces it; no simplicity of the first excited eigenvalue is assumed.

Finite-volume positivity supplies no positive volume-uniform lower bound. In particular $c(g)$ need not be continuous, and no regularity of $\mathcal{G}$ follows just from Proposition 1.

4b. Spectral limits and surviving weight

If $c(g) = 0$, then $\delta(g; N_s, G) \to 0$ along a subsequence of volumes. This is a statement about eigenvalues of different operators, not about the spectrum of a constructed infinite-volume operator. To discuss correlation length, first assume a limiting vacuum representation of the local gauge-invariant observable algebra, with cyclic vacuum Ω, nonnegative dimensionless Hamiltonian h_∞, and $\ker h_\infty = \mathbb{C}\Omega$. The physical time generator is $H_\infty = (\hbar c/a)h_\infty$. Existence of this representation and convergence of the relevant correlators are extra hypotheses here.

For each bounded local gauge-invariant observable A, set $A_0 = A - \langle \Omega, A\Omega\rangle I$ and define its finite positive spectral measure and temporal connected autocorrelation by

$$\mu_A(B) = \langle A_0\Omega, \mathbf{1}_B(h_\infty)A_0\Omega\rangle, \qquad C_A(s) = \int_{[0,\infty)} e^{-s\lambda}\, d\mu_A(\lambda), \quad s = \frac{ct_E}{a}.$$

For $\mu_A \neq 0$ define $r_A = \inf \operatorname{supp} \mu_A$ and

$$r = \inf_{A:\mu_A\neq 0} r_A, \qquad \xi_t = \frac{a}{r},$$

with $\xi_t = \infty$ when $r = 0$. This takes the **slowest rate over all local gauge-invariant observables**, rather than the rate of an arbitrarily chosen observable. It is a temporal correlation length; no equality with a spatial correlation length at fixed Hamiltonian lattice spacing is asserted.

Proposition 2 (conditional replacement). In the vacuum representation just specified, with a nonzero excited subspace,

$$m_{\mathrm{lat}} := \inf \sigma(h_\infty|_{\Omega^\perp}) = r, \qquad \Delta_\infty = \frac{\hbar c}{a} m_{\mathrm{lat}} = \frac{\hbar c}{\xi_t}.$$

For every nonzero μ_A, $-\lim_{s\to\infty} s^{-1} \log C_A(s) = r_A$. These assertions concern the assumed limit theory; they make no claim that $m_{\mathrm{lat}} = c(g)$.

Proof. Positivity of the spectral measure gives $C_A(s) \le \mu_A([0,\infty))e^{-sr_A}$. For every $\varepsilon > 0$, the definition of support gives positive weight $w_\varepsilon = \mu_A([r_A, r_A + \varepsilon]) > 0$, whence $C_A(s) \ge w_\varepsilon e^{-s(r_A+\varepsilon)}$. Taking logarithms, then $s \to \infty$ and $\varepsilon \downarrow 0$, proves the rate formula. Centered local vectors are dense in $\Omega^\perp$ by cyclicity of Ω for the local algebra. All their spectral supports lie above m_{lat}, so $r \ge m_{\mathrm{lat}}$. For every $\varepsilon > 0$ the spectral projection on $[m_{\mathrm{lat}}, m_{\mathrm{lat}} + \varepsilon)$ in $\Omega^\perp$ is nonzero. Density implies that it has nonzero action on some centered local vector. That vector has $r_A < m_{\mathrm{lat}} + \varepsilon$, proving the reverse inequality after $\varepsilon \downarrow 0$. The physical-unit identity follows from the definition of H_∞. □

An infimum of rates equal to zero need not be attained by a single observable: $\xi_t = \infty$ need not mean that *one* observable has slower-than-exponential decay. It can mean that there is no common positive exponential rate for the entire local algebra.

Why the original implication fails. Even convergence of an observable's spectral measures does not preserve their lowest support points. With d_x denoting unit point mass at x, consider

$$\mu_n = (1 - e^{-n})\mathrm{d}_1 + e^{-n}\mathrm{d}_{1/n}.$$

Its threshold tends to zero, but $\mu_n \Rightarrow \mathrm{d}_1$, whose threshold is one. The soft mode loses all its observable weight. This is a spectral-measure illustration, not a counterexample constructed for the fixed Kogut–Susskind family. It identifies information that finite-volume eigenvalues alone lack. Nonlocal sectors and the full-space/physical-space distinction add further obstacles.

A valid sufficient implication is available **with surviving weight**. Suppose centered finite-volume local spectral measures $\mu_n \Rightarrow \mu_A$ in the representation above and, for every $\eta > 0$, there is $w_\eta > 0$ with $\mu_n([0, \eta/2]) \ge w_\eta$ eventually. The closed-set inequality gives $\mu_A([0, \eta/2]) \ge w_\eta$. Uniqueness of the vacuum and centering give $\mu_A(\{0\}) = 0$, so $\mu_A((0,\eta)) > 0$ and $m_{\mathrm{lat}} = 0$.

These are precisely the extra convergence and weight hypotheses of three continuum limits, §5 (reading: passage). No such estimates are proved here for arbitrary g.

Correction (2026-10-02): phase interpretation. Vacuum coexistence with small finite-volume tunnelling splittings and a gapless local phase are possible scenarios, not an exhaustive consequence of $c(g) = 0$. Distinct infinite-volume ground states may persist throughout a symmetry-broken interval, with a positive excitation gap in each selected vacuum representation. A massless phase can likewise occupy an interval. Neither every degeneracy nor every gapless coupling is an isolated first- or second-order transition; neither an exponential volume law for tunnelling nor transition order was proved by the original argument.

Corollary 3 (original equivalence withdrawn, 2026-10-02). There is no proved equivalence here between $\mathcal{G} = (0, \infty)$ and absence of bulk phase transitions. T2′ means only the all-coupling lower-bound property defined in the lead. Identifying it with phase behaviour needs a limit theory, a specified vacuum/sector and spectral comparison.

Precision, 2026-09-23, corrected 2026-10-02. The openings note, §1 (reading: passage) records Wilson-action numerical evidence for first-order bulk transitions for $SU(N \geq 5)$, with source Rindlisbacher–Rummukainen–Salami. It does not prove a hole in the $\mathcal{G}$ of this KS Hamiltonian: the Euclidean Wilson operator differs, and the spectral comparison is missing. Such intermediate-coupling lattice phenomena do not by themselves obstruct a continuum limit along $g \to 0$. All-coupling T2′ is stronger than the weak-trajectory gap and scaling control needed for T3, together with continuum construction and nontriviality T4.

4c. What a second-order point could mean

Correction (2026-10-02). A divergent correlation length is a necessary scaling signal for a continuum limit with a fixed nonzero physical mass, not a proof that a continuum theory exists. The original assertion that a second-order point automatically produces a continuum limit, and the equivalence with uniqueness of the asymptotically free limit, are withdrawn.

If a positive dimensionless local spectral mass $m_{\mathrm{lat}}(g)$ tends to zero as $g \to g_* > 0$, one can *choose* $a(g) = m_{\mathrm{lat}}(g)/M$ to keep the inverse-length scale $M > 0$ fixed. Then $\Delta(g) = \hbar c M$ and $\xi_t/a = 1/m_{\mathrm{lat}} \to \infty$. This choice supplies scale matching only. Convergence of renormalized correlators, positivity, axioms and nontriviality remain to be proved. Moreover, diverging length alone does not specify transition order.

An asymptotically free trajectory instead approaches $g = 0$ with $a\Lambda_{\mathrm{lat}}(g) \to 0$ as described in §5 below. A possible finite-g_* scaling limit would be a different trajectory; T2′ alone classifies neither continuum limits nor universality classes. Numerically, the fundamental–adjoint plane of $SU(3)$ exhibits scalar softening consistent with a critical endpoint and a ϕ^4 continuum interpretation (Heller 1995, reading: abstract). This is evidence for that interpretation, not a constructed scalar continuum theory or a theorem for the Wilson axis.

4d. The abelian case: action, phase and transition order

The strong-coupling theorem applies to compact $U(1)$: in the normalization $-\Delta$ has eigenvalues n^2, so $C_2(R_{\min}) = 1$ and the defining representation has dimension one. The T2 note (reading: full-read) gives the KS bound

$$\delta(g; N_s, U(1)) \geq \gamma \frac{g^2}{2}, \qquad g^4 \geq \frac{24}{\beta_*},$$

with existential constants $\gamma, \beta_* > 0$ independent of volume. This proves a strong-coupling half-line is contained in $\mathcal{G}$, not that $\mathcal{G}$ is exactly a half-line.

For four-dimensional Euclidean compact $U(1)$, Guth proves a nonconfining weak-coupling phase for the **Villain** action (reading: abstract); Fröhlich–Spencer establishes a massless Coulomb phase in the setting recorded in B78 (reading: abstract). Those phase results do not

determine transition order or a continuously vanishing confined-phase mass. They also do not establish the exact set $\mathcal{G}_{U(1)} = (g_c, \infty)$ for the KS operator defined here; that translation would need its own proof.

For the standard four-dimensional **Wilson** action, numerical evidence supports a **weakly first-order** zero-temperature bulk transition (Vettorazzo–de Forcrand 2004, §§2–3, reading: passage). This replaces the erroneous second-order assertion. The Maxwell description belongs to the weak-coupling massless phase; no Maxwell continuum construction at the transition endpoint is established here. A massless phase over an interval also defeats the original identification of every gapless coupling with a transition point.

The abelian comparison remains useful: group-independent strong-coupling gap arguments alone do not supply weak-side control. The commutator potential of G07, §3, the local-observable restriction of the upper-bound note, Proposition 6, and the formal valley calculation of the valley note, §3b are distinct ingredients with distinct scopes. None is by itself a proof of a nonabelian infinite-volume or continuum gap, or of the global shape of either group's gapped set.

4e. Openness and closedness

For any subset of the connected interval $(0, \infty)$,

$$\mathcal{G} = (0, \infty) \quad \Longleftrightarrow \quad \mathcal{G} \neq \varnothing \text{ and } \mathcal{G} \text{ is open and closed.}$$

This is a topological fact, not a regularity theorem for $c(g)$.

property	status	what it would take
$\mathcal{G} \neq \varnothing$	proved by T2: $[g_0, \infty) \subset \mathcal{G}$	strong-coupling theorem
$\mathcal{G}$ open	unproved globally; (g_0, ∞) is a known interior region	volume-uniform stability under changes of g
$\mathcal{G}$ closed	unproved; sufficient for T2′ if openness holds	positivity at every finite limit point of gapped couplings, with control preventing lower bounds from collapsing

Correction (2026-10-02): later locality result. The original text left a Lieb–Robinson extension for the unbounded electric terms as the next obstacle. The later locality note, §§1–2 (reading: full-read) treats the commuting on-link electric terms in the interaction picture, leaving bounded finite-range plaquette interactions with unchanged support and norm. This addresses the locality ingredient; no numerical velocity or clustering-length estimate from that note is needed here. Its §5 also explains why locality alone does not prove gap stability: the cited stability route has additional hypotheses not verified at general finite KS coupling, including frustration-freeness. No universal assertion that every stability theorem requires that hypothesis is intended.

Closedness alone is not T2′ without openness, and excluding the two originally proposed scenarios would not establish either property. Finite-volume analytic dependence provides neither estimate. Even a proved T2′ would still leave continuum scaling, construction and nontriviality to T3/T4.

5. T3: the scaling limit, as one asymptotic statement

Along the continuum curve the bare coupling runs with the cutoff. With $b_0 = 11N/(48\pi^2)$ and $b_1 = 34N^2/(3(16\pi^2)^2)$ for $SU(N)$ (the two-loop coefficients in the convention $\beta(g) = -b_0 g^3 -$

$b_1 g^5 + \cdots$; Gross–Wilczek, Politzer, abstract, B78), the lattice Λ parameter is defined by

$$a\Lambda_{\rm lat}(g) = (b_0 g^2)^{-b_1/(2b_0^2)}\, e^{-1/(2b_0 g^2)}\,[1 + O(g^2)],$$

a dimensionless function that vanishes faster than any power of g.

T3. The limit $\dfrac{m}{\hbar c\,\Lambda_{\rm lat}} = \lim\limits_{g\to 0} \dfrac{\delta_\infty(g)}{a\Lambda_{\rm lat}(g)}$ exists and lies in $(0, \infty)$.

This is a proposed lattice-route scaling condition for a mass parameter m; identifying it with the gap of the reconstructed limit needs spectral transfer and surviving physical weight: the dimensionless infinite-volume gap must vanish as $g \to 0$ exactly like $e^{-1/(2b_0 g^2)}$ times the prescribed power, with a positive finite prefactor. Any weaker vanishing gives $m = \infty$ in the continuum (the theory is gapped only at the cutoff scale, as the compact abelian theory in three dimensions is, G07 §4); any faster vanishing, or $\delta_\infty = 0$ at arbitrarily small g, gives $m = 0$. T3 presupposes $\delta_\infty > 0$ on an interval $(0, g_1)$, which T2′ implies. The order of limits is $N \to \infty$ at fixed g first, then $g \to 0$ with $a = \Lambda_{\rm lat}^{-1}\, a\Lambda_{\rm lat}(g)$; Jaffe and Witten note (p. 6, passage) that a gap uniform in the volume would also serve to construct the infinite-volume limit, which is the reverse order.

The small-volume corner. For $L \ll \hbar c/\Lambda$ the constant-mode sector of the continuum torus theory has gap $\delta_1 g(L)^{2/3}\hbar c/L$ (C133, G07 Proposition 7), and Lüscher's expansion (Nucl. Phys. B219 (1983) 233, abstract) supplies the corrections in powers of $g^{2/3}$ with g the renormalized coupling at scale L. In the variable $z(L) = \Delta(L)L/(\hbar c)$ the two ends read

$$z(L) = \delta_1\, g(L)^{2/3}\,[1 + O(g^{2/3})] \quad (L \to 0), \qquad z(L) \sim \frac{mL}{\hbar c} \quad (L \to \infty),$$

and Lüscher–Münster (Nucl. Phys. B232 (1984) 445, abstract) place the crossover near $z \simeq 2$. The conjecture, in this variable, is that $z(L)$ is a function that starts at zero and grows without bound, with m the slope of its linear asymptote. The corner is the only place where the gap of the continuum theory is presently computable, and it is controlled by the mechanism of G08.

6. T4: existence

The lattice route needs the continuum limit of the Euclidean lattice theory with Osterwalder–Schrader reconstruction. Known: ultraviolet stability of four-dimensional lattice Yang–Mills in a finite volume with the cutoff removed (Balaban, Commun. Math. Phys. 122 (1989) 355 and the series it belongs to; Magnen–Rivasseau–Sénéor, Commun. Math. Phys. 155 (1993) 325, with an infrared cutoff; both metadata). Open: the infinite-volume limit, the axioms in infinite volume, and any spectral information. Jaffe and Witten (p. 7, passage) write that a construction on a compact space alone would be a major step and that no present ideas give a gap uniform in the volume.

7. The list

Statement	Content	Status
T1	finite lattice: unique physical ground state, gap $\Delta_{a,L} = (\hbar c/a)\delta(g; N_s, G) > 0$	proved (Theorem 1)

Statement	Content	Status
T2	$\delta(g; N_s, G) \geq c(N)g^2$ for $g \geq g_0(N)$, uniformly in N	established: Euclidean form (Osterwalder–Seiler); Hamiltonian form as a corollary of Yarotsky's theorem in the strong-coupling note
T2′	$\delta_\infty(g) > 0$ for all $g > 0$	open; the cited Euclidean U(1) phase does not by itself identify the KS gapped set
T3	$\delta_\infty(g)/(a\Lambda_{\mathrm{lat}}(g)) \to m/(\hbar c\Lambda_{\mathrm{lat}}) \in (0, \infty)$	open; spectral transfer to the reconstructed theory is also required
T4	continuum infinite-volume theory with the axioms	selected ultraviolet-stability inputs; construction and axioms open
S	small-volume corner $z(L) = \delta_1 g(L)^{2/3}[1 + O(g^{2/3})]$	C133 plus Lüscher's expansion

T2′ is a stronger all-coupling route, not a necessary independent condition for the continuum goal. T3 and T4 still require physical spectral transfer, nontriviality and the stated limit order.

Existing estimates and examined routes

For the $SU(3)$ Wilson transfer matrix, the separate sufficient bounds are $g^2 > 24/\log(19/18)$ (integer $g^2 > 444$ sufficient) from Dobrushin, and $g^2 \geq 6/\log(1 + 21/(500e^2))$ (integer 1059 sufficient) from the all-representation polymer proof. Their physical lower rates in units $\hbar c/a$ are respectively

$$\log \frac{1}{18(e^{24/g^2} - 1)}, \qquad 4 \log \frac{1}{176(e^{6/g^2} - 1)}.$$

The latter rate additionally needs $g^2 > 6/\log(177/176)$ to be positive. The leading-activity figure 176, linearization 1056 and sufficient integer 1059 are different statements. Neither criterion is a phase boundary. For KS, the explicit volume-uniform 388 threshold is withdrawn; the finite-volume correction is $\delta^{\mathrm{phys}} \geq (8/3)g^2 - 12|\mathcal{P}|/g^2$ under its stated lattice restrictions. The Yarotsky application retains existential volume uniformity. These are distinct operators and bounds.

The following are failures of the specified routes, not an exhaustion theorem:

route	why it closes
variational upper bounds, for $m > 0$	a spectral measure with all negative moments finite can have $m = 0$
expansion around the free theory	the bound reaches m only at the confinement scale, where its coefficient is nonperturbative
blocking by projection	a surface-to-gap estimate worsens by $3M^2/8$ in the stated strong-coupling asymptotic; all-coupling monotonicity is not proved
conjugation by the flow	the spectrum is invariant; a changed local decomposition requires control of generated terms
flow before decimation	sup norms are bijection-invariant and spectra conjugation-invariant

route	why it closes
Agmon estimate on the ground state	$\|\nabla V\|_\infty$ is extensive, so it bounds the global excess only
pointwise Gibbs domination	the comparison function bites only above the extensive threshold $V_* \simeq N\|\mathcal{P}\|$

The exact magnetic-energy sum rule and ground-state transfer remain useful identities. A normalized isotropic Wilson estimate does not transfer verbatim to the anisotropic KS ground state; projector convergence and matching of the measure/observable are needed. A flowed variational upper bound requires a finite derivative expectation and nonzero physical susceptibility; it is not by itself a positive lower gap.

The nonabelian matrix-model commutator valley, Gauss-law exclusion of a c-number linear electric probe and the formal torus-holonomy potential have different proof scopes (matrix gap, electric observables, valley). None fixes the shape of the KS gapped set or a four-dimensional continuum gap.

The gauge atlas owns the refinement-depth map. A local action cost and background count prove no normalized rarity theorem. A kernel tail proves no Hamiltonian relative comparison. The signed operator test diagnoses one localization/surface-loss construction, not every possible Hamiltonian method. The crossover comparison retains what the examined correlation and finite-certification tools would actually require.

8. Consequence for STATE

T2 in Hamiltonian form is now closed in the strong-coupling note as a corollary of Yarotsky's gap-stability theorem, with threshold $g_0(N) = (48N^2/[(N^2-1)\beta_*])^{1/4}$; the unresolved task is to control the weak-coupling scaling trajectory and transfer the physical spectrum to the reconstructed theory. The $d = 3$ target of STATE keeps its place as the case where T3 has no transmutation to perform.